

Website Traffic Analysis

Data Analytics Internship Project

Submitted By

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Project Objective

The objective of this project is to analyze website traffic and user interaction data to understand audience engagement, geographic traffic distribution, popular artists, and user activity trends using data analytics and visualization techniques.

Tools & Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Google Colab
-

Key Analysis Performed

- Data Cleaning & Preprocessing
 - Traffic Event Analysis
 - Geographic Traffic Analysis
 - Artist & Track Popularity Analysis
 - User Engagement Trends
 - Event Distribution Analysis
 - Business Recommendations
-

Expected Outcome

The analysis helps identify user engagement trends, audience behavior, and optimization opportunities for improving digital platform performance.

▼ Importing Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

✓ Loading the Dataset

```
df = pd.read_csv('/content/traffic.csv')
```

```
df.head()
```

	event	date	country	city	artist	album	track	isrc	
0	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	1c
1	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	1c
2	click	2021-08-21	India	Ludhiana	Reyanna Maria	So Pretty	So Pretty	USUM72100871	34
3	click	2021-08-21	France	Unknown	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más	BRUM72003904	08

✓ Understanding the Dataset

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   event       226278 non-null object
1   date        226278 non-null object
2   country     226267 non-null object
3   city        226267 non-null object
4   artist      226241 non-null object
```

```

5   album      226273 non-null object
6   track      226273 non-null object
7   isrc       219157 non-null object
8   linkid     226278 non-null object
dtypes: object(9)
memory usage: 15.5+ MB

```

```
df.describe(include='all')
```

	event	date	country	city	artist	album	track	
count	226278	226278	226267	226267	226241	226273	226273	219
unique	3	7	211	11993	2419	3254	3562	
top	pageview	2021-08-19	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070
freq	142015	35361	47334	22791	40841	40841	40841	40

```
df.isnull().sum()
```

	0
event	0
date	0
country	11
city	11
artist	37
album	5
track	5
isrc	7121
linkid	0

```
dtype: int64
```

✓ Data Cleaning & Preprocessing

```
df.drop_duplicates(inplace=True)
```

```
df['date'] = pd.to_datetime(df['date'])
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 122567 entries, 0 to 226274
Data columns (total 9 columns):
 #   Column      Non-Null Count  Dtype  
---  -
 0   event       122567 non-null  object 
 1   date        122567 non-null  datetime64[ns]
 2   country     122562 non-null  object 
 3   city        122562 non-null  object 
 4   artist      122539 non-null  object 
 5   album       122563 non-null  object 
 6   track       122563 non-null  object 
 7   isrc        116261 non-null  object 
 8   linkid      122567 non-null  object 
dtypes: datetime64[ns](1), object(8)
memory usage: 9.4+ MB
```

✓ KPI Analysis

✓ TOTAL EVENTS

```
total_events = df['event'].count()

print("Total Events:", total_events)
```

```
Total Events: 122567
```

✓ TOTAL COUNTRIES

```
total_countries = df['country'].nunique()

print("Total Countries:", total_countries)
```

```
Total Countries: 211
```

✓ TOTAL CITIES

```
total_cities = df['city'].nunique()

print("Total Cities:", total_cities)
```

```
Total Cities: 11993
```

✓ TOTAL ARTISTS

```
total_artists = df['artist'].nunique()

print("Total Artists:", total_artists)

Total Artists: 2419
```

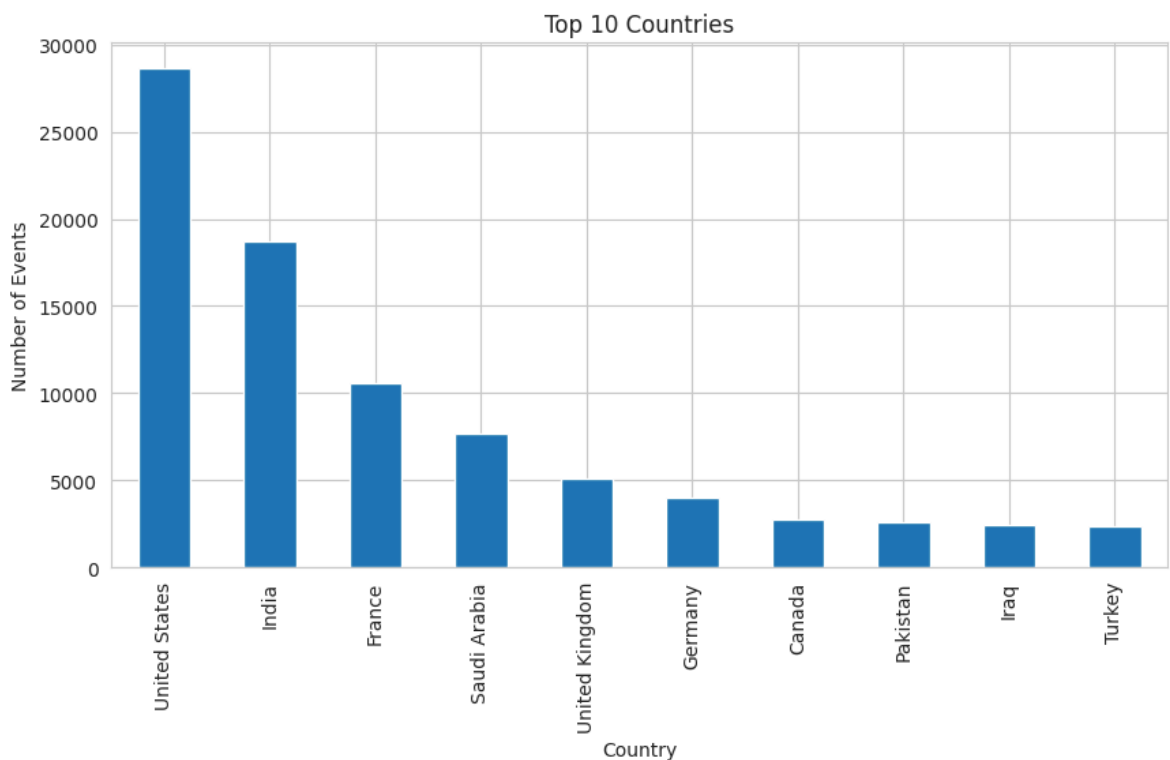
Insight

The KPI analysis provides an overview of total traffic events, geographic reach, artist engagement, and audience interaction patterns within the dataset.

✓ 🌍 Top Countries Analysis

```
top_countries = df['country'].value_counts().head(10)

top_countries.plot(kind='bar', figsize=(10,5))
plt.title('Top 10 Countries')
plt.xlabel('Country')
plt.ylabel('Number of Events')
plt.show()
```



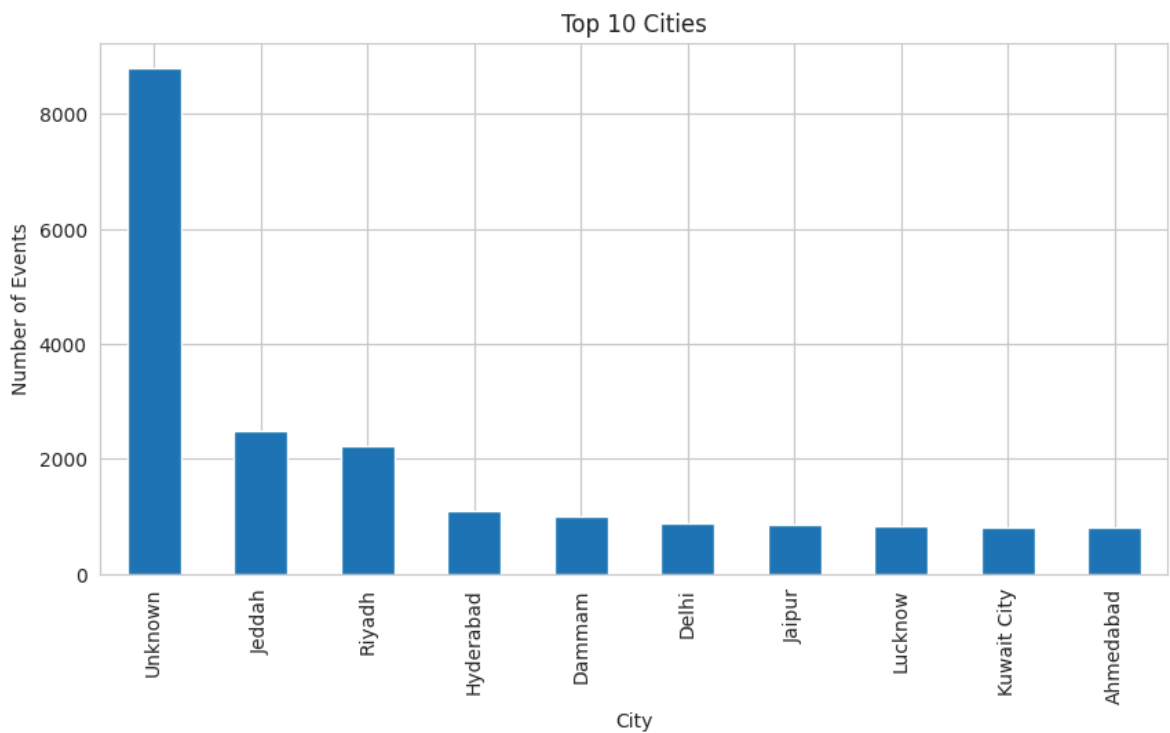
Business Insight

Certain countries generate significantly higher traffic activity, indicating stronger audience engagement and platform reach in those regions.

▼ Top Cities Analysis

```
top_cities = df['city'].value_counts().head(10)

top_cities.plot(kind='bar', figsize=(10,5))
plt.title('Top 10 Cities')
plt.xlabel('City')
plt.ylabel('Number of Events')
plt.show()
```



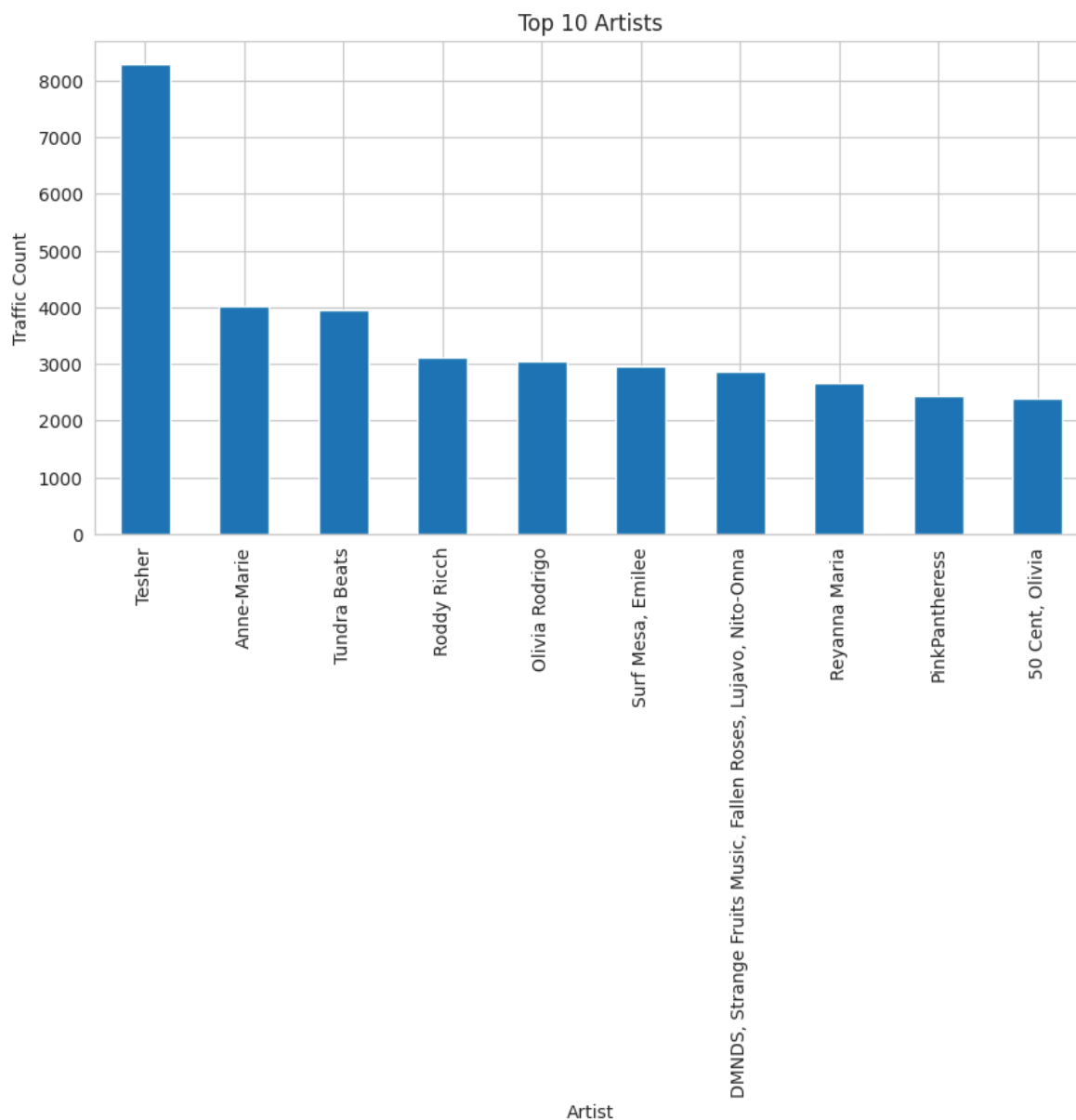
Business Insight

The analysis identifies cities with the highest user interaction and traffic activity, helping understand audience concentration.

Top Artists Analysis

```
top_artists = df['artist'].value_counts().head(10)

top_artists.plot(kind='bar', figsize=(10,5))
plt.title('Top 10 Artists')
plt.xlabel('Artist')
plt.ylabel('Traffic Count')
plt.show()
```



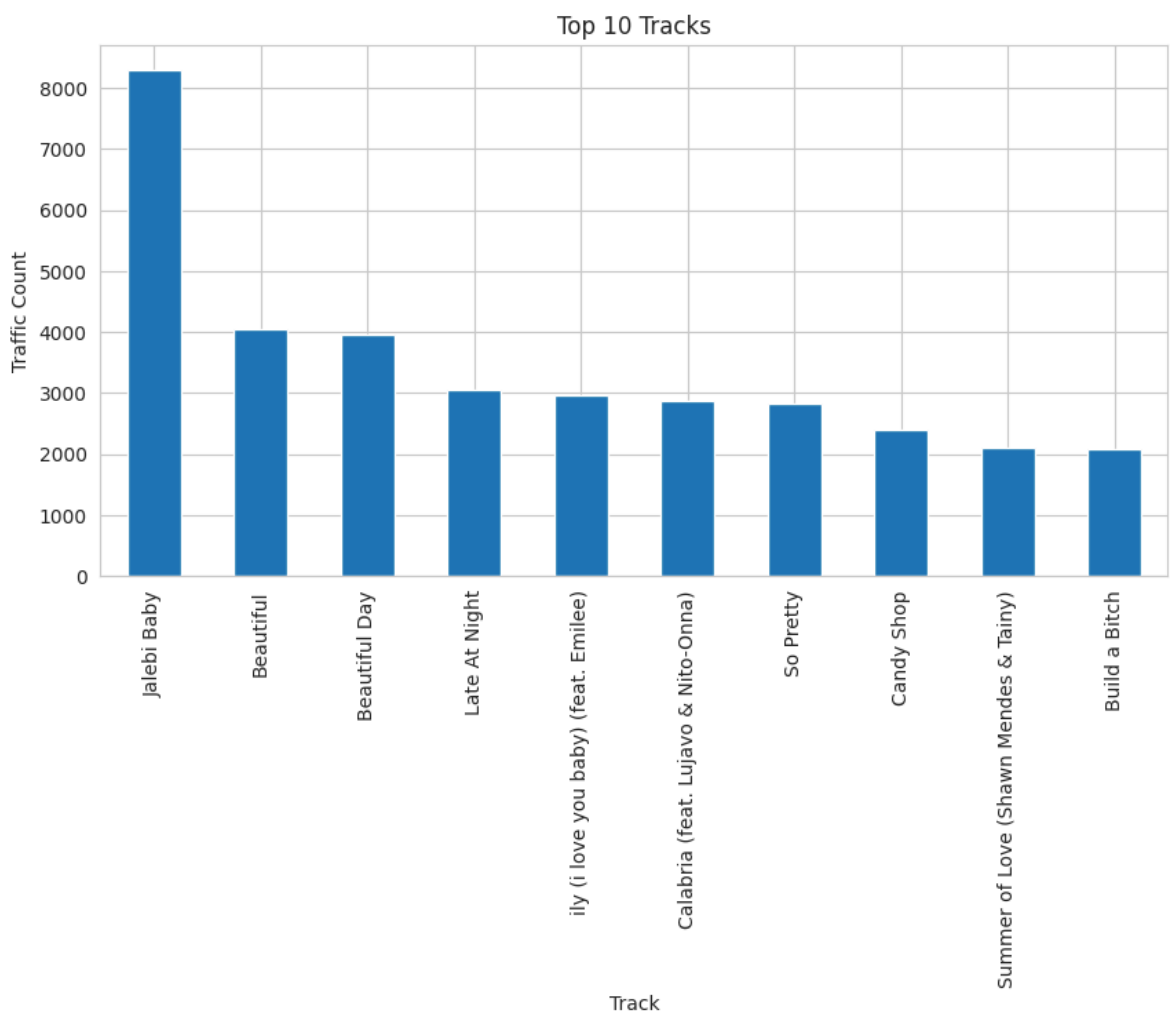
Business Insight

Popular artists attract higher user traffic and engagement, indicating strong audience interest and content popularity.

🎧 Top Tracks Analysis

```
top_tracks = df['track'].value_counts().head(10)
```

```
top_tracks.plot(kind='bar', figsize=(10,5))  
plt.title('Top 10 Tracks')  
plt.xlabel('Track')  
plt.ylabel('Traffic Count')  
plt.show()
```



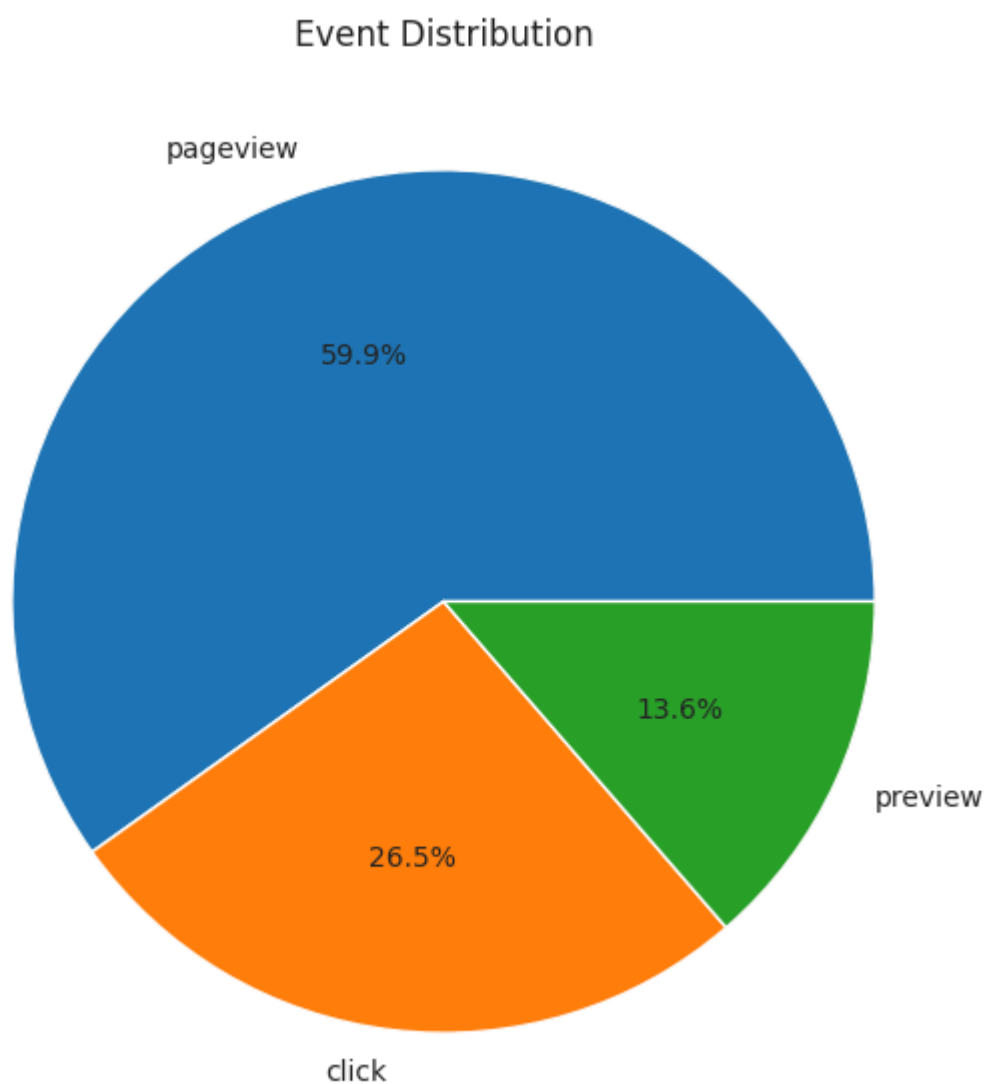
Business Insight

Certain tracks generate significantly higher engagement and user interaction, reflecting trending content and listener preferences.

✓ Event Distribution Analysis

```
event_distribution = df['event'].value_counts()

event_distribution.plot(kind='pie', autopct='%1.1f%%', figsize=(7,7))
plt.title('Event Distribution')
plt.ylabel('')
plt.show()
```



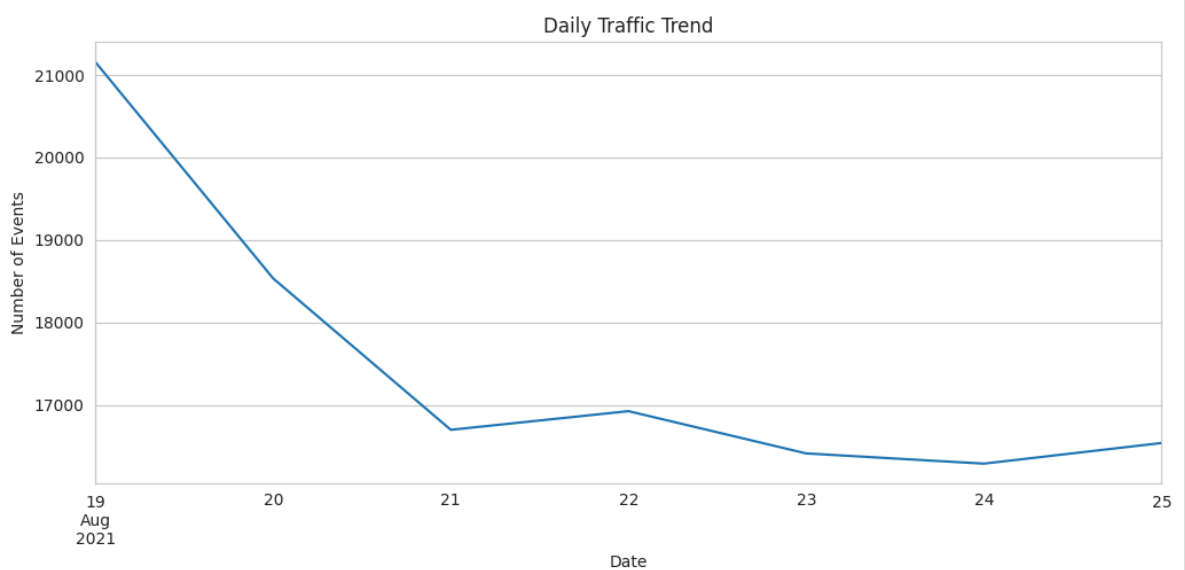
Business Insight

Event distribution analysis helps understand user activity behavior and interaction patterns on the platform.

✓ 📅 Daily Traffic Trend Analysis

```
daily_traffic = df.groupby('date')['event'].count()

plt.figure(figsize=(12,5))
daily_traffic.plot()
plt.title('Daily Traffic Trend')
plt.xlabel('Date')
plt.ylabel('Number of Events')
plt.show()
```

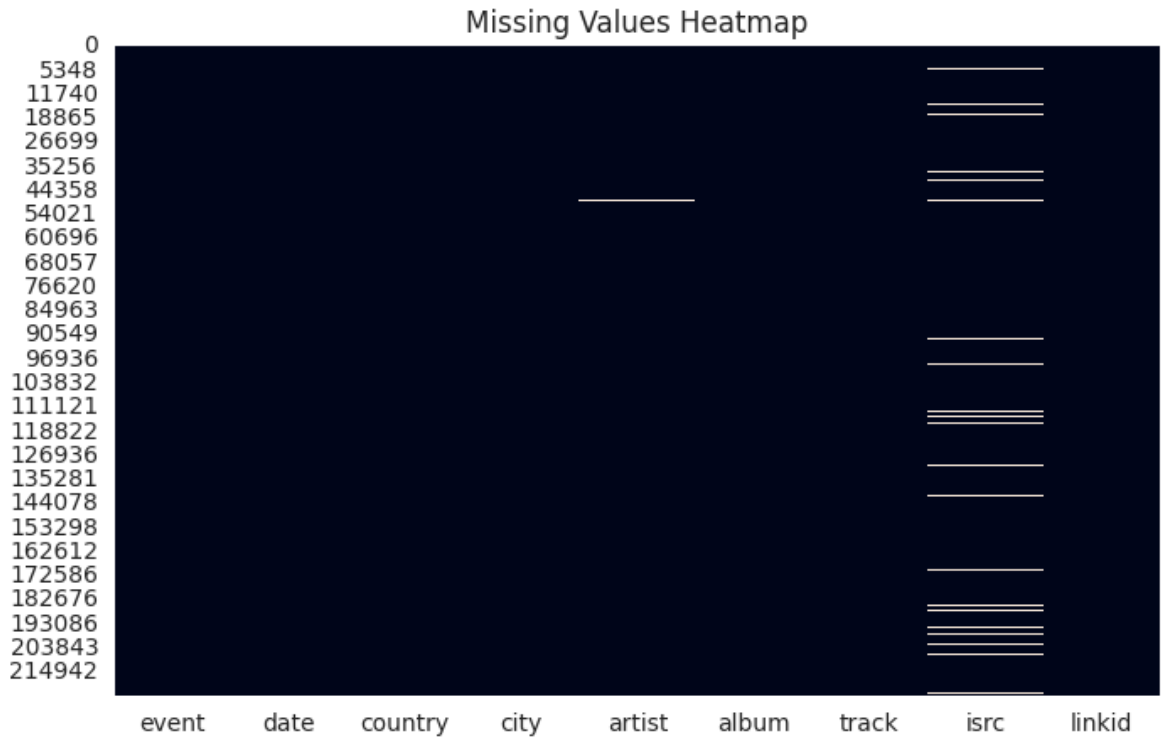


Business Insight

Traffic activity fluctuates over time, indicating changing user engagement patterns and audience interaction trends.

✓ 🔥 Missing Values Heatmap

```
plt.figure(figsize=(8,5))
sns.heatmap(df.isnull(), cbar=False)
plt.title('Missing Values Heatmap')
plt.show()
```



Business Insight

The heatmap helps visualize missing data patterns and improves understanding of dataset quality.



Business Recommendations

1. Focus marketing campaigns in high-traffic countries and cities.
2. Promote popular artists and tracks to improve engagement.
3. Analyze low-performing regions for audience growth opportunities.
4. Monitor daily traffic trends to optimize content release timing.
5. Use audience interaction data to improve recommendation systems and personalization strategies.

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☒ Final Conclusion

The Website Traffic Analysis project successfully identified important audience engagement trends, geographic traffic patterns, popular artists, and user interaction beh



Final Conclusion

popular artists, and user interaction behavior.

The analysis demonstrates how traffic and interaction data can support better digital marketing strategies, audience targeting, and platform optimization decisions.

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